

NBA Athlete Value Predictor

Liam Whitenack
School of Data Science & Analytics (M.S.)
Old Dominion University
Hampton, Virginia
lwhit073@odu.edu

Abstract—Each professional basketball player’s contract-based salary reflects their perceived value and can be viewed as a quantitative measure of their impact, skill, and contribution in the NBA. Accurately estimating salary therefore provides a mechanism for comparing and ranking players according to their theoretical value, offering a data-driven perspective on player quality. In this study, we develop a nonlinear framework that leverages performance metrics and related factors to estimate player salaries. By modeling the relationship between on-court performance and compensation, we construct a system that enables players to be evaluated on a common scale of inferred value. Additionally, we examine the relative importance of different features in shaping these estimates, providing further insight into the attributes most closely associated with high-value players. Our approach reframes salary prediction as a tool for player evaluation, allowing for the systematic ranking of athletes based on their projected worth. This perspective highlights how statistical performance translates into perceived value, offering a quantitative foundation for comparing players across the league.

I. INTRODUCTION

As shown in previous work [1].

REFERENCES

- [1] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.